[B19EC2101]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

II/IV B.Tech I Semester (R19) Regular Examinations

ELECTRONIC DEVICES AND CIRCUITS

Common to ECE & EEE

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

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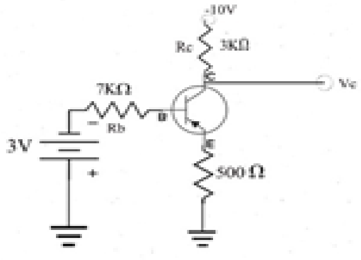
			CO	KL	M
		UNIT-I			
1.	**a).**	Discuss in detail the various current components in a p-n junction diode.	**CO1**	**K2**	**7**
	b).	A Silicon diode operates at a forward voltage of 0.4 V. Calculate the factor by which the current will be multiplied when the temperature is increased from 25 degree centigrade to 150 degree centigrade.	**CO1**	**K3**	**8**
		OR			
2.	**a).**	With a neat diagram, explain the operation of a full wave rectifier and obtain expressions for ripple factor and efficiency.	**CO1**	**K4**	**7**
	b).	Distinguish between Zener breakdown and Avalanche breakdown.	**CO1**	**K5**	**8**
		UNIT-II			
3.	**a).**	Explain input and output characteristics of the transistor in CE configuration with a neat sketch.	**CO1**	**K2**	**7**
	b).	Compare CE, CC and CB configurations of a transistor.	**CO1**	**K5**	**8**
		OR			
4.	**a).**	Explain the early effect and its consequences in the Common base transistor.	**CO1**	**K2**	**7**
	b).	 For the circuit shown, β_{dc} = 100 and V_{BE} = 0.8V. (i). Find if the transistor is in cut off, saturation or active region. (ii). Find the voltage (V_c). (iii). Find the minimum value of R_B for which the transistor operates in an active region.	**CO1**	**K3**	**8**
		UNIT-III			
5.	**a).**	Explain how self-bias circuit improves stability of operating point and obtain an expression for the stability factor 'S' for the self-bias circuit	**CO2**	**K4**	**7**

	b).	Explain how diodes can be used to compensate against variations in V_{BE} and I_{CO} due to change in temperature.	CO2	K2	8
		OR			
6.	a).	A transistor with β= 100 is to be used in Common Emitter Configuration with collector to base bias. R_E = 1 KΩ, V_{CC} = 10V. Assume V_{BE} = 0.7V. Choose R_B so that the quiescent collector to emitter voltage is 4V. Find Stability Factor S.	CO2	K3	7
	b).	Explain thermal runaway and thermal stability in transistors.	CO2	K2	8
		UNIT-IV			
7.	a).	Compare JFET and BJT.	CO1	K5	7
	b).	Draw the basic structure of N- channel JFET and explain the operation with the help of characteristic curves.	CO1	K2	8
		OR			
8.	a).	Explain the construction and operation of Enhancement MOSFET with drain and transfer characteristics.	CO1	K2	7
	b).	A JFET is to be connected in a circuit with a drain load resistor of 4.7 kΩ, and a supply voltage of V_{DD} = 30V. V_D is to be approximately 20V, and is to remain constant within ± 1 V. Design a suitable self-bias circuit.	CO2	K3	8
		UNIT-V			
9.	a).	Derive expressions for current gain, Voltage gain, Input impedance and Output impedance of a generalized transistor amplifier using h-parameters.	CO3	K4	7
	b).	Explain the frequency response of the CE amplifier with a neat sketch.	CO3	K2	8
		OR			
10.	a).	Discuss the effect of bypass and coupling capacitors in CE amplifiers.	CO3	K2	7
	b).	Draw and explain a hybrid **π-** model of a transistor used at high frequencies.	CO3	K2	8

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

[B19EC2102]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

II/IV B.Tech I Semester (R19) Regular Examinations

SWITCHING THEORY AND LOGIC DESIGN

Electronics and Communication Engineering

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

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			CO	KL	M
		UNIT-I			
1.	**a).**	Convert 1096_{10} into Binary, Octal, Hexadecimal, BCD and Excess-3.	1	3	8
	b).	Simplify the following functions by using basic Boolean laws i) $\bar{X}\bar{Z}+XYZ+X\bar{Z}=Y$ ii) $A\bar{B}C+(\bar{B}+\bar{C})(\bar{B}+\bar{D})+\overline{(A+C+D)}=Y$	1	2	7
		OR			
2.	**a).**	Explain how you can detect single error occurred during transmission in a 7 bit Hamming code word with an example.	1	3	8
	b).	i) Prove that $(A\oplus B)\oplus C = A\oplus(B\oplus C)$ ii) Simplify $\bar{A}B + ABD + A\bar{B}CD + BC = Y$	1	2	7
		UNIT-II			
3.	**a).**	Express the following in SOP and POS forms $i)\ (AB+C)(B+\bar{C}D)$ $ii)\ \bar{x}+x(x+\bar{y})(y+\bar{z})$	2	2	6
	b).	Simplify using K-Map f(A,B,C,D) = ∑m(0,1,3,7,8,12)+∑d(5,10,13,14) and draw the corresponding circuit with only NAND gates.	2	3	9
		OR			
4.	**a).**	Simplify the given function by using K-map f(A,B,C,D) =∑m(0,1,4,5,7,11,15). + ∑d (10, 13)	2	2	6
	b).	Find Prime Implicants of f(w,x,y,z) = =∑m(1,4,6,7,8,9,10,11,15) by using Quine – McCluskey method.	2	3	9
		UNIT-III			
5.	**a).**	Design an 8:1 Multiplexer with only NAND Gates.	3	4	8
	b).	Implement the given functions by using a 3 to 8 Decoder and logic Gates. (i) f_1 (A,B,C) = Σ m(0,1,5,7) (ii) f_2 (A,B,C) = Σ m(1,3,5,7)	3	3	7
		OR			
6.	**a).**	Design and draw a BCD to 7 segment Decoder circuit for active	3	4	8

		low outputs.			
	b).	What is the function of priority encoder? Design a 3bit priority encoder.	3	3	7
		UNIT-IV			
7.	**a).**	Convert D-FF into JK-FF.	3	4	8
	b).	What is a race around problem in a JK flip-flop and explain the method to avoid it by using Master Slave JK-FF.	3	3	7
		OR			
8.	**a).**	Draw a 3-bit right shift register and explain the modes SIPO, SISO, PIPO and PISO.	3	3	7
	b).	Design Mod-5 Ring counter and draw its output waveforms.	3	4	8
		UNIT-V			
9.	**a).**	Explain difference between synchronous and asynchronous circuits.	4	3	7
	b).	Design a Sequence detector which produces an output 0 every time the sequence 0101 is detected, an output 1 at all other times. Assume that overlapping sequence is allowed. Use D-FFs in your design.	4	4	8
		OR			
10.	**a).**	Design a synchronous sequential circuit using D-FFs for the given state table.	4	4	8

Present State		Next State				Output	
		X=0		X=1		X=0	X=1
A	B	A	B	A	B	A	B
0	0	0	0	0	1	0	0
0	1	0	0	1	1	1	0
1	0	0	0	1	0	1	0
1	1	0	0	1	0	1	0

	b).	What is lockout in counters? Design a Mod-10 synchronous counter by using D-Flip- Flops.	4	3	7

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

[B19EC2103]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE(A)

II/IV B.Tech I Semester (R19) Regular Examinations

SIGNALS AND SYSTEMS

ELECTRONICS AND COMMUNICATION DEPARTMENT

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

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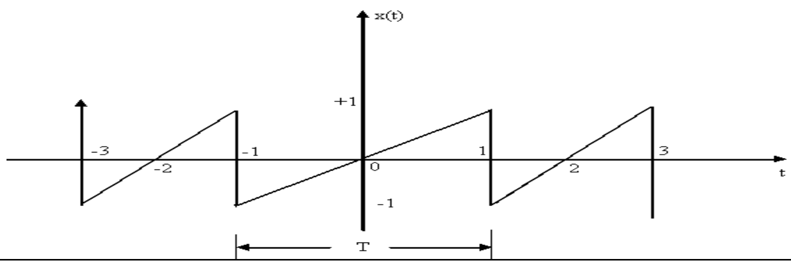
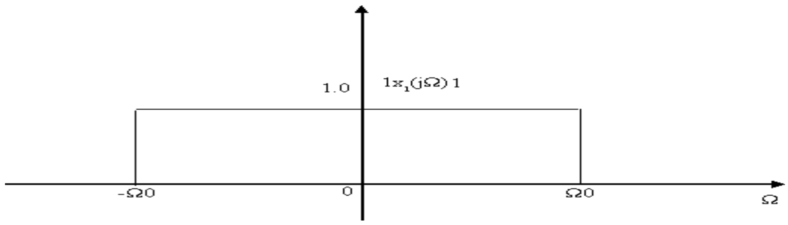
			CO	KL	M
		UNIT-I			
1.	**a).**	Explain all classification of signals with examples for each category.	**1**	**2**	**8**
	b).	Determine the power and RMS value of the following signals 1 . $x\ (t) = 5\cos\ (50t + 3)$ 2 . $x\ (t) = 10cos(5t)cos(10t)$	**1**	**3**	**7**
		OR			
2.	**a).**	Determine whether the following systems are time invariant or not. 1 . $y\ (t) = \mathrm{x}(t)$ 2 . $y\ (n) = x(2n)$	**1**	**3**	**8**
	b).	Prove the energy of the power signal is infinite over infinite time.	**1**	**2**	**7**
		UNIT-II			
3.	**a).**	Find the trigonometric Fourier series for the periodic signal x(t) as shown in figure below 	**2**	**3**	**7**
	b).	State and prove parseval"s relation of Fourier series.	**2**	**2**	**8**
		OR			
4.	**a).**	State and prove the time shifting and time scaling properties of CTFS.	**2**	**2**	**7**
	b).	Explain Dirichlet conditions on Convergence of Fourier Series.	**2**	**2**	**8**
		UNIT-III			
5.	**a).**	State and prove the differentiation in the frequency domain property of CTFT.	**3**	**2**	**8**
	b).	Find x(t) from X(jw) $=1/(1+jw)^2$	**3**	**3**	**7**
		OR			

6.	**a).**	Find the inverse Fourier transform of X(jΩ) for the spectra shown in figure below.  	**3**	**3**	**7**
	b).	Find the Fourier transform of the following: 1. $e^{at}u(-t)$ 2. $te^{-at}u(t)$	**3**	**3**	**8**
		UNIT-IV			
7.	**a).**	State and prove the final value theorem of Laplace transform.	**4**	**2**	**8**
	b).	Compute the initial and final values for $X(S)= (2S+5)/(S(S+3)(S+4)^2)$	**4**	**2**	**7**
		OR			
8.	**a).**	Explain ROC properties of Laplace transforms.	**4**	**2**	**8**
	b).	Determine the bilateral Laplace transform and ROC for the following signals. (i) $e^{-2t}\,u(t-2)$ (ii) $e^{-4\lvert t\rvert}$	**4**	**3**	**7**
		UNIT-V			
9.	**a).**	Find the inverse z-transform of the following: 1. $X(z) = \frac{1}{(1+3Z^{-1}+2Z^{-2})}$ ROC: mod(Z) >2 2. $X(z) = \frac{(Z+1)}{(4+35+Z^2)}$ ROC: mod(Z)<2	**5**	**3**	**7**
	b).	Solve the following difference equation for y(n) using z-transform for the specified Initial condition. $y(n) - y(n-1) + \frac{1}{4}\big(y(n-2)\big) = X(n) : n > 0$ $where\ X(n) = 2\left(\frac{1}{8}\right)^n ; y(-1) = 2;\ y(-2) = 4.$	**5**	**3**	**8**
		OR			
10.	**a).**	A signal x(t)=sin c(150π t) is sampled at a rate of (i) 100Hz (ii) 200 Hz (iii) 300Hz. For each of these cases, explain if you can recover the signal x(t) from the sampled signal.	**6**	**3**	**7**
	b).	Find the z-transform and ROC for the following: 1. $x(n)= a^n\ u(n)$ 2. $x(n)= -b^n\ u(-n-1)$.	**5**	**3**	**8**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

[B19EC2104]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

II/IV B.Tech I Semester (R19) Regular Examinations

PROBABILITY THEORY & RANDOM PROCESSES

Electronics and Communication Engineering

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

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			CO	KL	M
		UNIT-I			
1.	**a).**	State and prove Bayes theorem.	**CO1**	**K2**	**8**
	b).	A single card is drawn from a 52 card deck (i) What is the probability that the card is a jack? (ii) What is the probability that the card will be a 5 (or) smaller? (iii) What is the probability that the card is a red 10?	**CO1**	**K3**	**7**
		OR			
2.	**a).**	State and prove Addition theorem of probability.	**CO1**	**K2**	**8**
	b).	In a classroom, 70% are above average, 20% are average and 10% are below average. Suppose that 20% of above average, 10% of average and 20% of below average students fail in a subject. What is the probability that a randomly selected student is an average student who failed?	**CO1**	**K3**	**7**
		UNIT-II			
3.	**a).**	Define Distribution and Density functions. List the properties of them. Prove any one property of each.	**CO2**	**K2**	**7**
	b).	A random variable X has a probability density function $f_X(x) = \frac{c}{x^2+1}$ $-\alpha < X < \alpha$ Find the constant 'c' and also find the probability distribution function of the random variable X.	**CO2**	**K3**	**8**
		OR			
4.	**a).**	The joint probability density function of two random variables x and y is given by $f_{XY}(x,y) = e^{-(x+y)}$ If $x \geq 0, y \geq 0$, = 0 otherwise. Find the marginal density functions of x and y. Also prove that x and y are independent.	**CO2**	**K3**	**8**
	b).	A random variable X has a Gaussian density function with mean '0'and variance '1'. Find the probability density function of the random variable Y defined as $Y = X2$.	**CO2**	**K3**	**7**
		UNIT-III			

5.	**a).**	State and prove Chebyshev's inequality .	**CO3**	**K2**	**7**
	b).	A random variable has a probability density function $f_X(\boldsymbol{x}) \ =\frac{5}{4}(1-x^4) \quad for\ 0 < x \leq 1$, $= 0$ elsewhere Find (i) E(X) (ii) E (4X+2) (iii) E(X^2).	**CO3**	**K3**	**8**
		OR			
6.	**a).**	Compute the characteristic function for a random variable X with f_X(**x**)= $\frac{1}{2}\ e^{-\lvert x\rvert}$, $-\alpha \leq x \leq \alpha$	**CO3**	**K3**	**7**
	b).	State and Prove Central Limit Theorem.	**CO3**	**K2**	**8**
		UNIT-IV			
7.	**a).**	Define Autocorrelation and power spectral density for a random process. List their Properties.	**CO4**	**K2**	**8**
	b).	Determine the PSD of a stationary process whose autocorrelation is given by $Rxx(\tau)=\sigma^2 e^{-\alpha\lvert\tau\rvert}$	**CO4**	**K3**	**7**
		OR			
8.	**a).**	Prove that the random process is wide sense stationary $X(t)=A\cos(\omega_0 t+)$ where A,ω_0 are constants and is uniformly distributed random variable on the interval $(0,2\pi)$.	**CO4**	**K3**	**8**
	b).	Distinguish between stationary and Ergodic random processes.	**CO4**	**K3**	**7**
		UNIT-V			
9.	**a).**	Find the input autocorrelation function, output autocorrelation function and output spectral density of RC low pass filter, when the filter is subjected to a white noise of spectral density No/2.	**CO5**	**K5**	**8**
	b).	Show that the Narrow band noise process can be expressed as In-phase and Quadrature components.	**CO5**	**K2**	**7**
		OR			
10.	**a).**	What is Low pass and Band pass noise representation and write their properties?	**CO5**	**K2**	**7**
	b).	Write short notes on (i) Poisson Process (ii) Gaussian Process	**CO5**	**K2**	**8**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

[B19EE2105]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

II B. Tech I Semester (R19) Regular Examinations

NETWORK ANALYSIS

Electronics and Communication Engineering

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

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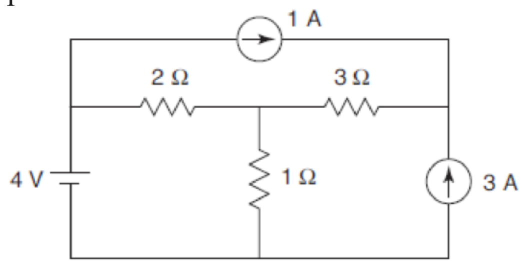
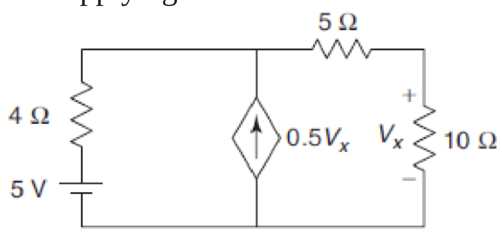
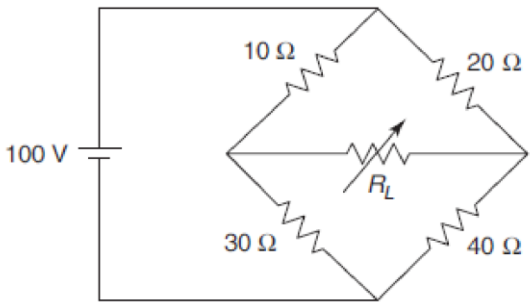
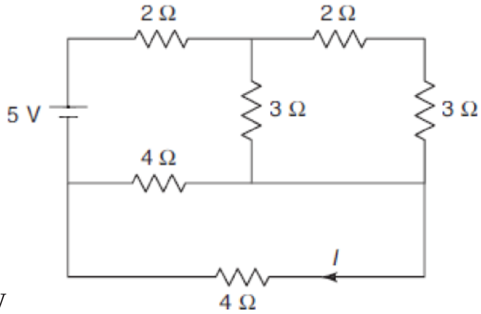
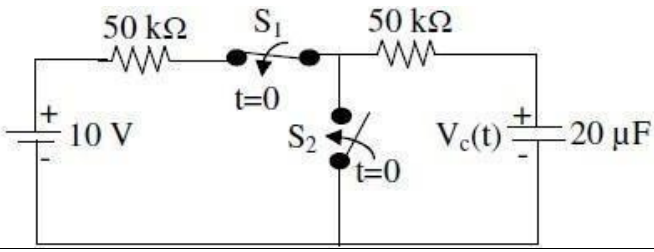
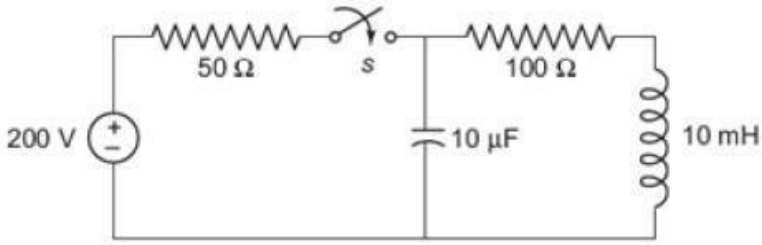
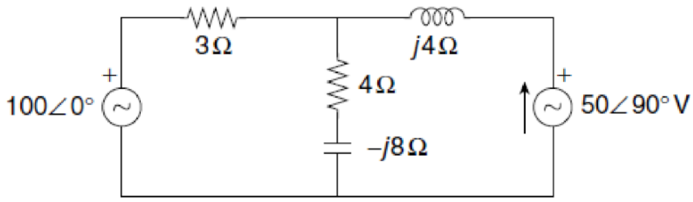
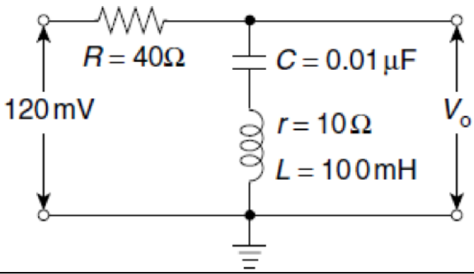
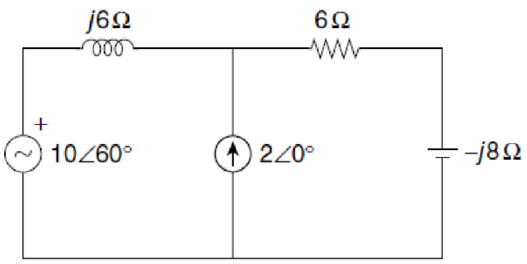
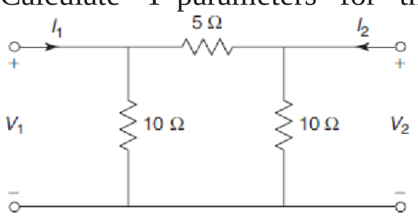
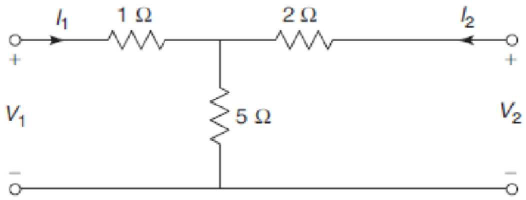
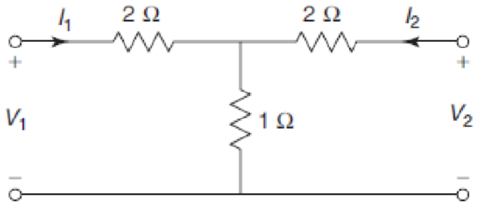
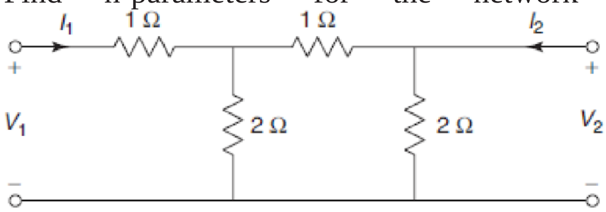
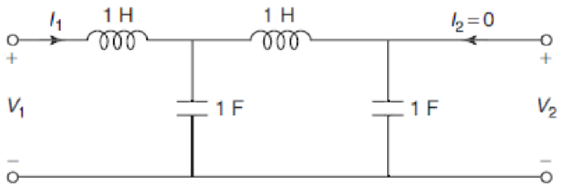
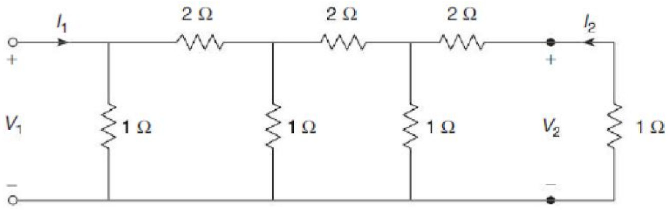
			CO	PO	M
		UNIT-I			
1.	**a).**	Apply Super position theorem to find the current in the 1Ω  resistor.	**CO1**	**PO1**	**7**
	b).	Find the current through the 10 Ω resistor for the network by applying thevenin's theorem. 	**CO1**	**PO1**	**8**
		OR			
2.	**a).**	Find the value of resistance R_L in the given network for maximum power transfer and calculate the maximum power by applying maximum power transfer principle. 	**CO1**	**PO1**	**7**

	b).	Find the current I and verify reciprocity theorem for the network  shown below	**CO1**	**PO1**	**8**
		UNIT-II			
3.	**a).**	Derive an expression for the current in an RC circuit excited by a unit step voltage. And draw the corresponding plot.	**CO2**	**PO1**	**7**
	b).	In the circuit of figure, switch S_1 has been closed for a long time while switch S_2 has been open for a long time. At t=0, switch S_1 opens and switch S_2 closes. Determine the value of voltage Vc(t) for all t>0. 	**CO2**	**PO2**	**7**
		OR			
4.	**a).**	Derive an expression for the current in an RL circuit excited by a unit step voltage.	**CO2**	**PO1**	**8**
	b).	When the switch is closed at $t = 0$, determine the transient current through inductor for the network shown in below Figure. Assume that initial current through the inductor is zero. 	**CO2**	**PO2**	**7**
		UNIT-III			
5.	**a).**	Determine the current through $(4 - j8)$ Ω branch in the circuit shown in Figure using Thevenin's theorem and Norton's theorem.	**CO3**	**PO1**	

					
	b).	An *R–L–C* circuit has the following values: *R* =40Ω, *C*=0.01 μF, *L* =100 mH and the resistance of the coil is 10 Ω. An input voltage 120 mV of variable frequency is appearing at the input. The output is taken out from the *L–C* combination. Calculate the output voltage at resonance. Further, calculate the bandwidth. 	**CO3**	**PO1**	8
		OR			
6.	**a).**	Find the current through 6Ω resistor shown in Figure by applying mesh analysis. 	**CO3**	**PO1**	7
	b).	A series RLC circuit with R = 10 ohms, L = 0.4 H and C = 50 µF has applied voltage of 200V with variable frequency. Calculate the resonant frequency, current at resonance, voltage across R, L and C. Also calculate the Q-factor, upper and lower half power frequencies and bandwidth.	**CO3**	**PO1**	8
		UNIT-IV			
7.	**a).**	Calculate Y-parameters for the network shown in below. 	**CO4**	**PO1**	**7**

	b).	Two identical sections of the network shown in Fig., are connected in cascade. Obtain the transmission parameters of the overall connection.  	**CO4**	**PO1**	**7**
		OR			
8.	**a).**	Two identical sections of the network shown in below fig., are connected in series. Obtain Z-parameters of the overall connection.  	**CO4**	**PO1**	**7**
	b).	Find h-parameters for the network shown below.  	**CO4**	**PO1**	**8**
		UNIT-V			
9.	**a).**	Find the network functions V_2/V_1 , V_2/I_1 and V_2/I_1 for the network shown in below  fig.,	**CO5**	**PO1**	**7**
	b).	Find the function v(t) using the pole-zero plot of following function $V(s) = \dfrac{(s+2)(s+6)}{(s+1)(s+5)}$	**CO5**	**PO1**	**7**
		OR			

10.	**a).**	For the below resistive two-port network find V_2/V_1 , V_2/I_1, I_2/I_1  and I_2/V_1	**CO5**	**PO1**	**8**
	b).	The denominator of a network function is expressed as a polynomial For what value of *K*, the system will be stable? $Q(s) = s^3 + 2s^2 + 3s + K$	**CO5**	**PO2**	**7**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

[B19CS2108]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

II B. Tech I Semester (R19) Regular Examinations

DATA STRUCTURES

COMMON TO ECE & EEE

MODEL QUESTION PAPER

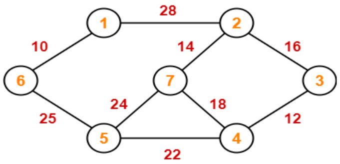
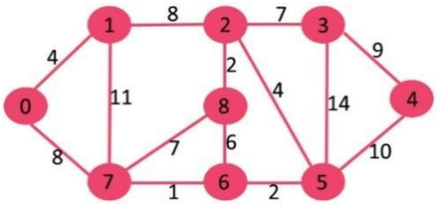
TIME: 3Hrs. **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT-I				**CO-PO**	**K**
1.	**(a)**	**Applying** stack operations Write an algorithm for postfix evaluation and **Applying** Postfix Evaluation algorithm Evaluate the following postfix expression. 2 3 8 5 1 + - $ * 9 6 - *	**8M**	**CO1 -PO1**	**K3**
	(b)	**Construct** a non-recursive algorithm for binary search. Write Non-Recursive algorithms for finding the n^{th} Fibonacci number.	**7M**	**CO1 -PO1**	**K3**
(OR)					
2.	**(a).**	**Identify** C routines for primitive operations on stack	**8M**	**CO1 -PO1**	**K3**
	(b).	**Applying** stack operations Write an algorithm for infix to postfix. Apply algorithm Convert the following infix expression to postfix expression.A + (B * C - (D / E $ F) * G) * H	**7M**	**CO1 -PO1**	**K3**
UNIT-II					
3.	**(a).**	**Apply** Linked list to perform stack operations.	**8M**	**CO2 -PO1**	**K3**
	(b).	**Apply** Double Linked list to perform insertion operation.	**7M**	**CO2 -PO1**	**K3**
(OR)					
4.		**Build** an algorithm to find transpose of the given sparse matrix and also implement it by using a c program.	**15 M**	**CO2 -PO1**	**K3**
UNIT-III					
5.		Explain Binary tree traversal techniques and **Construct** a Binary Tree from **inorder 4,8,2,5,1,6,3,7** and **post order 8,4,5,2,6,7,3,1.**	**15 M**	**CO3 -PO1**	**K3**
(OR)					
6.	**(a).**	**Construct** a Threaded binary tree by taking Binary Tree example	**8M**	**CO3 -PO1**	**K3**
	(b).	**Construct** Min Heap an Max Heap for given Input 35 33 42 10 14 19 27 44 26 31	**7M**	**CO-PO1**	**K3**
UNIT -IV					
7.	**(a).**	Explain warshall's algorithm and **identify** the transitive closure of a graph with an example.	**8M**	**CO3 -PO1**	**K3**

	(b).	**Apply** BFS traversal of a graph with an example	7M	CO3-PO1	K3
(OR)					
8.	(a)	**Construct** Minimum Spanning Tree by Kruskal algorithm 	8M	CO3-PO1	K3
	(b)	**Construct** Minimum Spanning Tree by Prims Algoritm 	7M	CO3-PO1	K3
UNIT-V					
9.		**Analyse** and sort the given input elements using quick sort technique 52, 37, 63, 14, 17, 8, 6, 25	15 M	CO4-PO2	K4
(OR)					
10.		**Analyse** with binary search technique to identify the element 65 in the give list of elements: 10,20,40,65,75,87,99,	15 M	CO4-PO2	K4

[B19EC2201]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

II/IV B.Tech I Semester (R19) Regular Examinations

ELECTRONIC CIRCUIT ANALYSIS

Department of Electronics and Communication Engineering

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

			CO	KL	M
		UNIT-I			
1.	**a).**	Show that the low band gain of an FET RC coupled amplifier at low frequency is $(-gmR_0R_g/(R_0^1+ R_g +1/j\omega C_c)$	**3**	**4**	**8M**
	b).	A transistor is connected as a CE amplifier with load resistance of 10KΩ. The parameters are h_{ie}=5KΩ and h_{fe}=330. Calculate the overall gain for mid frequency range when four such stages are connected in cascaded RC coupling. Assume R_s=0.	**2**	**3**	**7M**
		OR			
2.	**a).**	Explain what you understand by the term bandwidth shrinkage factor as applied to cascaded RC stages. Determine the expression for the same.	**2**	**3**	**7M**
	b).	Determine the expression for short circuit current Gain-Bandwidth product of a CE amplifier.	**2**	**3**	**8M**
		UNIT-II			
3.	**a).**	Determine expressions for input impedance and output impedance in case of a voltage series feedback topology.	**2**	**3**	**8M**
	b).	An amplifier gain change by ±10% using negative feedback amplifier is to be modified to yield gain of 100 with 0.1% variation. Calculate the required loop gain and amount of negative feedback.	**2**	**3**	**7M**
		OR			
4.	**a).**	Determine the expressions for input impedance and output impedance for a current shunt feedback topology.	**2**	**3**	**8M**
	b).	Evaluate the voltage gain, input impedance and output impedance of an amplifier with feedback for voltage series feedback having A= 100, RI=10Kohms, RO=20 Kohms and β = 0.1.	**3**	**4**	**7M**
		UNIT-III			
5.	**a).**	Determine the expression for the frequency of oscillations of an RC- Phase shift oscillator.	**2**	**3**	**7M**

	b).	In a Colpitts oscillator = 0.001 mF and = 0.01 mF and L = 5 mH. Evaluate (i).Frequency of oscillations (ii) If 'L' is doubled, find the new frequency.	**3**	**4**	**8M**
		OR			
6.	**a).**	Discuss about Crystal oscillators with neat diagrams and necessary equations.	**1**	**2**	**6M**
	b).	Demonstrate how the oscillations are built up in the Hartley oscillator with a neat circuit diagram and also derive the expression for the same.	**2**	**3**	**9M**
		UNIT-IV			
7.	**a).**	Compare single tuned and stagger tuned amplifiers.	**1**	**2**	**6M**
	b).	Classify the Power amplifiers and Show that the efficiency of a transformer coupled class A power amplifier is 50%	**2**	**3**	**9M**
		OR			
8.	**a).**	Sketch and explain the neat circuit diagram of a Single tuned amplifier and deduce the expression for gain.	**2**	**3**	**8M**
	b).	Explain how even harmonic distortion is reduced in a Class-A push-pull power amplifier.	**1**	**2**	**7M**
		UNIT-V			
9.	**a).**	With the help of neat diagrams, explain the following applications using Op-Amp. (i) Non-Inverting amplifier. (ii) Summing amplifier	**1**	**2**	**8M**
	b).	Explain the concept of "Virtual ground" for OP-AMPs and derive an expression for closed loop gain of inverting configuration of OP-AMP	**2**	**3**	**7M**
		OR			
10.	**a).**	Discuss about the following OP-AMP parameters: (i) CMRR, (ii) PSRR and (iii) I/P Bias Current	**1**	**2**	**7M**
	b).	Analyze and design an OP-AMP circuit to give an output $V_o = 3/4\ V_1 + 5/6\ V_2 + 6/7\ V_3$, where $V_1=V_2=V_3=1V$	**3**	**4**	**8M**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

[B19EC2202]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

II/IV B.Tech II Semester (R19) Regular Examinations

ELECTROMAGNETIC WAVES AND TRANSMISSION LINES

Electronics and Communication Engineering

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

			CO	KL	M
		UNIT-I			
1.	**a).**	A charge Q=−10nc is at the origin in free space. If the x−component of **E** is to be zero at the point (3,1,1), determine the value Q_t which should be kept at the point (2,0,0).	**1**	**K4**	**7**
	b).	State Gauss's law. Deduce the expression for Electric field due to an infinite line charge.	**1**	**K3**	8
		OR			
2.	**a).**	Define conduction current density. Derive the expression for continuity equation.	**1**	**K3**	8
	b).	The dielectric medium1 has permittivity $\varepsilon_{r1} = 2$, dielectric medium2 has permittivity $\varepsilon_{r2} = 4$ and electric field in medium1 is $\mathbf{E}_1=5\mathbf{a}_x + 3\mathbf{a}_y - 13/z\mathbf{a}_z$ v/m. Determine electric field in medium2 $\mathbf{E}_2$.	**1**	**K4**	**7**
		UNIT-II			
3.	**a).**	Explain about Biot Savart's law in detail.	**1**	**K3**	**7**
	b).	The Magnetic field intensity in a region of space is $\mathbf{H}=(x+2y)/z^2\mathbf{a}_y + 2/z\mathbf{a}_z$ A/M. Determine **J** and also the total current passing through the surface Z=4, 1≤X≤2, 3≤Y≤5.	**1**	**K4**	8
		OR			
4.	**a).**	Determine the magnetic potential at a distant point due to a small circular loop of radius a carrying a current I amperes.	**1**	**K4**	**7**
	b).	Explain about the Force due to the magnetic field on a moving charge and on a current element.	**1**	**K3**	8
		UNIT-III			

5.	**a).**	Deduce the expressions for Maxwell's equations in both differential and integral form.	**2**	**K3**	8
	b).	Given **E**=10sin ($\omega t-\beta z$)$\mathbf{a}_y$ V/m in free space, determine **D**, **B** and **H**.	**2**	**K4**	7
		OR			
6.	**a).**	Deduce the boundary conditions for **E**, **H**, **D** and **B** at the interface between two different dielectric media.	**2**	**K3**	8
	b).	A medium is characterized by $\sigma = 0$, $\mu = 2\mu_0$, and $\varepsilon = 5\varepsilon_0$. If **H** = 2cos {($\omega t$ — 3y)} $\mathbf{a}_z$, A/m, calculate ω and **E**.	**2**	**K4**	**7**
		UNIT-IV			
7.	**a).**	Define uniform plane wave. Deduce the expression for intrinsic impedance of free space.	**3**	**K3**	8
	b).	State wave propagation characteristics of good conductors. A medium is defined by $\sigma = 5.8 \times 10^7$ mho/m, $\varepsilon_r = 1$, $\mu_r = 1$, supports a uniform plane wave of frequency 60Hz. Determine the attenuation constant, propagation constant, intrinsic impedance, wavelength and phase velocity.	**3**	**K4**	**7**
		OR			
8.	**a).**	List different types of polarization. Explain about circular polarization of a uniform plane wave.	**3**	**K3**	**7**
	b).	Derive the expression for reflection and transmission coefficients, when a uniform plane wave incident normally on the surface of a perfect dielectric.	**3**	**K3**	8
		UNIT-V			
9.	**a).**	A distortionless line has Z_0= 50Ω, α = 20m Np/m, V =0.6C, where C is the speed of light in vacuum. Compute R, L, G, C and λ at 100 MHz.	**4**	**K3**	**7**
	b).	Derive the input impedance of a Transmission line in terms of characteristic impedance.	**4**	**K3**	8
		OR			
10.	**a).**	A rectangular waveguide has cross-sectional area of 2.29*1.45cm^2 and operating frequency is 10 GHz. Calculate (a) free space wavelength (b) cut-off wavelength (c) cut-off frequency (d) angle of incidence (e) guide wavelength (f) phase velocity (g) phase drift constant and wave impedances of the waveguide (TE and TM).	**4**	**K3**	8

	b).	Explain the reason for inability of rectangular waveguide to allow TEM mode propagation.	**4**	**K3**	**7**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

[B19 EC2203]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

II/IV B.Tech II Semester (R19) Regular Examinations

ANALOG COMMUNICATIONS

Department of Electronics and Communication Engineering

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 70 M**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

			CO	KL	M
		UNIT-I			
1.	**a)**	Explain the AM diode detector demodulator with a neat diagram.	**CO1**	**K2**	**7**
	b)	Explain the generation of DSB-SC with the Balanced modulator method.	**CO1**	**K2**	**7**
		OR			
2.	**a)**	Explain advantages of SSB system. Describe a method of generating an SSB signal.	**CO1**	**K2**	**7**
	b)	Explain Vestigial Side Band Modulation.	**CO1**	**K2**	**7**
		UNIT-II			
3.	**a)**	Define frequency deviation and phase deviation in Frequency Modulation and differentiate between NBFM and WBFM.	**CO2**	**K3**	**7**
	b)	In a FM system when the audio frequency is 500Hz and the AF voltage is 2.4 V, the frequency deviation is 4.8 kHz. If the AF voltage is increased to 7.2V, what is the new deviation? If the AF voltage is raised to 10V, while audio frequency is dropped to 200Hz, what is the deviation? Find the modulation index in each case.	**CO2**	**K3**	**7**
		OR			
4.	**a)**	Describe the operation of a phase locked loop FM demodulator with a neat diagram.	**CO2**	**K2**	**7**
	b)	Draw the block diagram of an Indirect method of FM generation and explain.	**CO2**	**K2**	**7**
		UNIT-III			
5.	**a)**	Derive the expression for signal -to-noise ratio in DSB-SC system.	**CO3**	**K3**	**8**
	b)	Explain sources of Noise.	**CO3**	**K2**	**6**
		OR			
6.	**a)**	Derive the expression for signal -to-noise ratio in FM systems.	**CO3**	**K3**	**10**

	b)	Explain Threshold in FM.	**CO3**	**K2**	**4**
		UNIT-IV			
7.	**a)**	Draw the block Diagram of AM transmitter and Explain.	**CO4**	**K2**	**8**
	b)	Explain Pre-Emphasis & De-emphasis.	**CO4**	**K2**	**6**
		OR			
8.	**a)**	Draw the block diagram of FM transmitter and Explain.	**CO4**	**K2**	**7**
	b)	Draw the block diagram of an SSB transmitter and Explain.	**CO4**	**K2**	**7**
		UNIT-V			
9.	**a)**	Explain the principle of operation of a super heterodyne radio receiver.	**CO5**	**K2**	**8**
	b)	What are the factors that govern the choice of Intermediate frequency?	**CO5**	**K2**	**6**
		OR			
10	**a)**	Distinguish between AGC and Delayed AGC with help of neat circuit diagrams.	**CO5**	**K4**	**7**
	b)	Define Image frequency and Image frequency Rejection Ratio.	**CO5**	**K2**	**7**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

[B19EC2204]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

II/IV B.Tech II Semester (R19) Regular Examinations

COMPUTER ARCHITECTURE AND ORGANIZATION

Electronics and Communication Engineering

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

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			CO	KL	M
		UNIT-I			
1.	**a).**	Explain the operation of a 4-bit adder-subtractor with a neat diagram.	**CO 1**	**K 3**	**7**
	b).	Design a common bus system with multiplexers to connect four 4 bit registers A,B,C,D. How this can be modified with the help of a three state buffer.	**CO 1**	**K 3**	**8**
		OR			
2.	**a).**	Explain in detail the operation of one stage arithmetic logic shift unit by using a function table for each input.	**CO 1**	**K 3**	**8**
	b).	The adder- subtractor circuit has following values for input mode M and data inputs A and B. In each case, determine the values of the outputs: S_3, S_2, S_1, S_0 and C_4. <table><tr><td>M</td><td>A</td><td>B</td></tr><tr><td>0</td><td>0111</td><td>0110</td></tr><tr><td>0</td><td>1000</td><td>1001</td></tr><tr><td>1</td><td>1100</td><td>1000</td></tr><tr><td>1</td><td>0101</td><td>1010</td></tr><tr><td>1</td><td>0000</td><td>0001</td></tr></table>	**CO 1**	**K 3**	**7**
		UNIT-II			
3.	**a).**	Explain about Timing and Control Unit design.	**CO 2**	**K 2**	**7**
	b).	With the help of a flowchart explain the operation of a basic computer during 1. Instruction cycle 2. Interrupt cycle	**CO 2**	**K 3**	**8**
		OR			
4.	**a).**	Explain in detail about basic instruction formats with examples.	**CO 2**	**K 3**	**7**
	b).	With the help of a diagram explain the functions of the basic computer registers connected to a common bus.	**CO 2**	**K 3**	**8**
		UNIT-III			

5.	a).	Write the merits and demerits of a micro programmed control unit compared to a hardwired control unit.	CO 3	K 2	7
	b).	Explain the concept of microprogramming in detail with the help of an example.	CO 3	K 4	8
		OR			
6.	a).	With an example, explain various addressing modes and write how to calculate the address in each case.	CO 3	K 4	8
	b).	What are the characteristics of a RISC processor? What is a CISC processor?	CO 3	K 2	7
		UNIT-IV			
7.	a).	Explain Daisy Chain Interrupt Mechanism. List the advantages and disadvantages in this scheme.	CO 4	K 2	7
	b).	With the help of block diagram and truth table explain the operation of an I/O interface unit`	CO 4	K 3	8
		OR			
8.	a).	Show internal configuration of DMA and explain its operation in a computer for read and write modes.	CO 4	K 3	8
	b).	Explain the operation of Asynchronous Communication Interface.	CO 4	K 2	7
		UNIT-V			
9.	a).	A system employs RAM chips of 256bytes and ROM chips of 1024bytes. It needs 2KB RAM, 4KB ROM and four interface units with four registers each and memory mapped I/O used. The two highest order bits of the address bus are 00 for RAM, 01 for ROM and 10 for interface registers. Draw a memory address map for the system	CO 4	K 3	8
	b).	What is Cache memory and explain two mapping methods in implementation of Cache.	CO 4	K 3	7
		OR			
10.	a).	Explain Memory hierarchy and Differentiate between Main memory and Auxiliary memories	CO 4	K 2	7
	b).	Explain the following 1. Associative Memory 2. Virtual Memory	CO 4	K 2	8

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARKS

[B19CS2209]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

II/IV B.Tech II Semester (R19) Regular Examinations

OOPS THROUGH JAVA

Common to ECE & EEE

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:75**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

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				CO	K
UNIT-I					
1.	**(a).**	Define an Array and explain different types of arrays. By **applying** the concept of an array find the largest element from a given set of elements in an array.	**15 M**	**CO1 -PO1**	**K3**
(OR)					
2.	**(a).**	Illustrate the differences between C, C++ and Java with a neat diagram	**8M**	**CO1 -PO1**	**K3**
	(b).	Illustrate the structure of a java program	**7M**	**CO1 -PO1**	**K3**
UNIT-II					
3.	**(a).**	Illustrate the concept of Inheritance and its different types with neat pictures.	**8M**	**CO2 -PO2**	**K3**
	(b).	Construct a sample java program of user choice which applies the functionality of inheritance.	**7M**	**CO2 -PO2**	**K3**
(OR)					
4.		Explain polymorphism and its types. Construct a java program which Illustrates the functionality of method overloading and method overriding.	**15 M**	**CO2 -PO2**	**K3**
UNIT-III					
5.		Illustrate how to solve the problem of multiple inheritance in java with an example.Also differentiate between class and an interface.	**15 M**	**CO3 -PO2**	**K3**
(OR)					
6.	**(a).**	Interpret the concept of packages in java.	**7M**	**CO3 -PO2**	**K3**
	(b).	Construct a java program that shows the functionality of creating a public class in an already existing user defined package.	**8M**	**CO3 -PO3**	**K3**
UNIT -IV					
7.	**(a).**	Compare **throw** and **throws** in Exception Handling	**8M**	**CO3 -PO2**	**K3**
	(b).	Construct a java program which creates a user defined exception	**7M**	**CO3 -PO3**	**K3**
(OR)					
8.		Identify the different ways of creating a Thread in java programming, show with examples programs wherever necessary.	**15 M**	**CO3 -PO3**	**K3**

UNIT-V					
9.		Illustrate Life cycle of an applet. Construct a sample Applet program which displays current date.	15 M	CO4 -PO3	K3
(OR)					
10.		Apply the concept of Event Handling and construct a java program which contains a button with name “day”,when clicked on it displays the current day.	15 M	CO4 -PO3	K3

[B19CS2209]

[B19HS2201]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

II/IV B.Tech II Semester (R19) Regular Examinations

MANAGEMENT AND ORGANIZATIONAL BEHAVIOR

Common to ECE & EEE

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

		CO	KL	M
	UNIT-I			
1.	Define Management and Explain its functions.	**CO1**	**K2**	**15**
	OR			
2.	Explain the principles of Management as outlined by Henry Fayol.	**CO1**	**K2**	**15**
	UNIT-II			
3.	Describe the functions performed by Human Resource Manager.	**CO2**	**K2**	**15**
	OR			
4.	Define Marketing, Explain in detail about Marketing mix.	**CO2**	**K2**	**15**
	UNIT-III			
5.	Explain about the importance of Mission, Goal, Objective and Strategy.	**CO3**	**K2**	**15**
	OR			
6.	What do you understand by SWOT analysis? Explain how it can be carried out.	**CO3**	**K2**	**15**
	UNIT-IV			
7.	What is Organisational Change and describe about the types of change.	**CO4**	**K2**	**15**
	OR			
8.	What is Motivation and Explain about Maslows Human Need Theory.	**CO4**	**K2**	**15**
	UNIT-V			
9.	Explain about the consequences of conflicts in an organisation.	**CO5**	**K2**	**15**
	OR			
10.	What is Stress & Describe about methods of managing Stress.	**CO5**	**K2**	**15**

[B19HS2201]